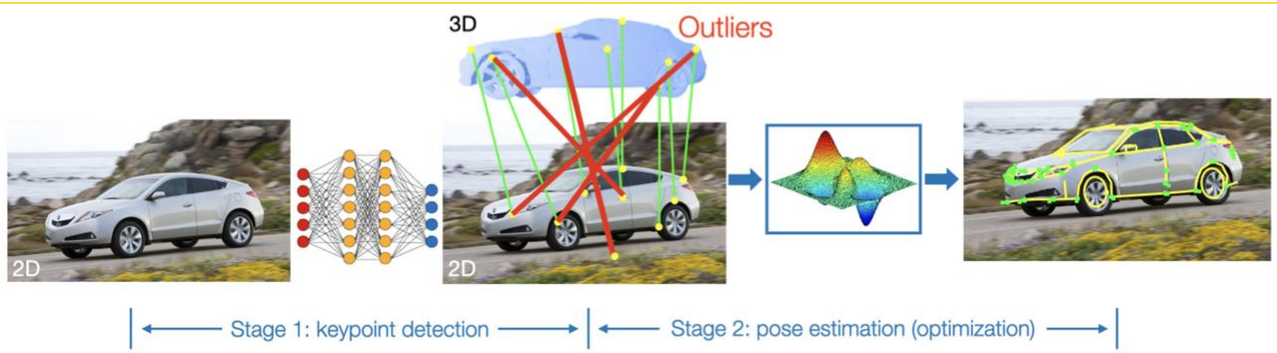
Conformal Semantic Keypoint Detection with Statistical Guarantees

Heng Yang and Marco Pavone

Workshop on Robot Learning: Trustworthy Robotics

Object Pose Estimation: why untrustworthy?



Keypoints are not guaranteed to be close to groundtruth

Focus of our paper

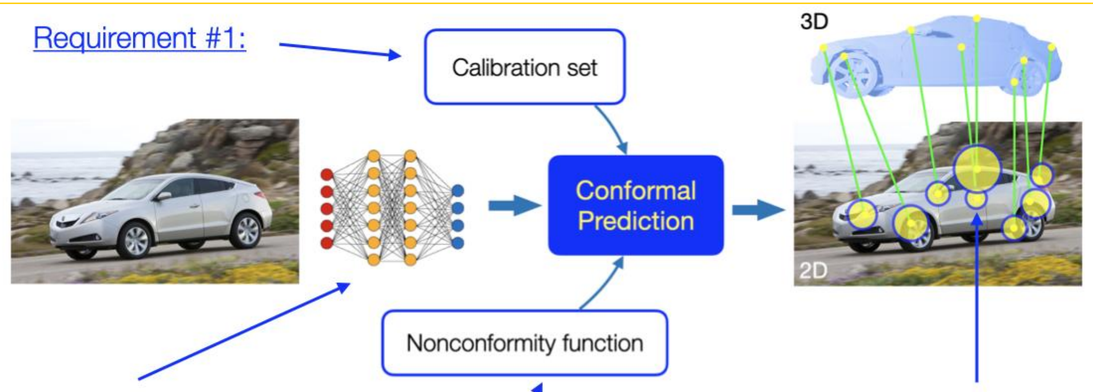
Popular heuristics (e.g., RANSAC) get trapped in local minima and fail without notice

Related work: certifiable algorithms [YC22TPAMI]

Our method: conformal keypoint detection

Paradigm shift: from heuristic prediction to probabilistically correct prediction sets

Key ingredient: inductive conformal prediction (ICP) — finite-sample uncertainty quantification without assumptions on the data distribution or the prediction model

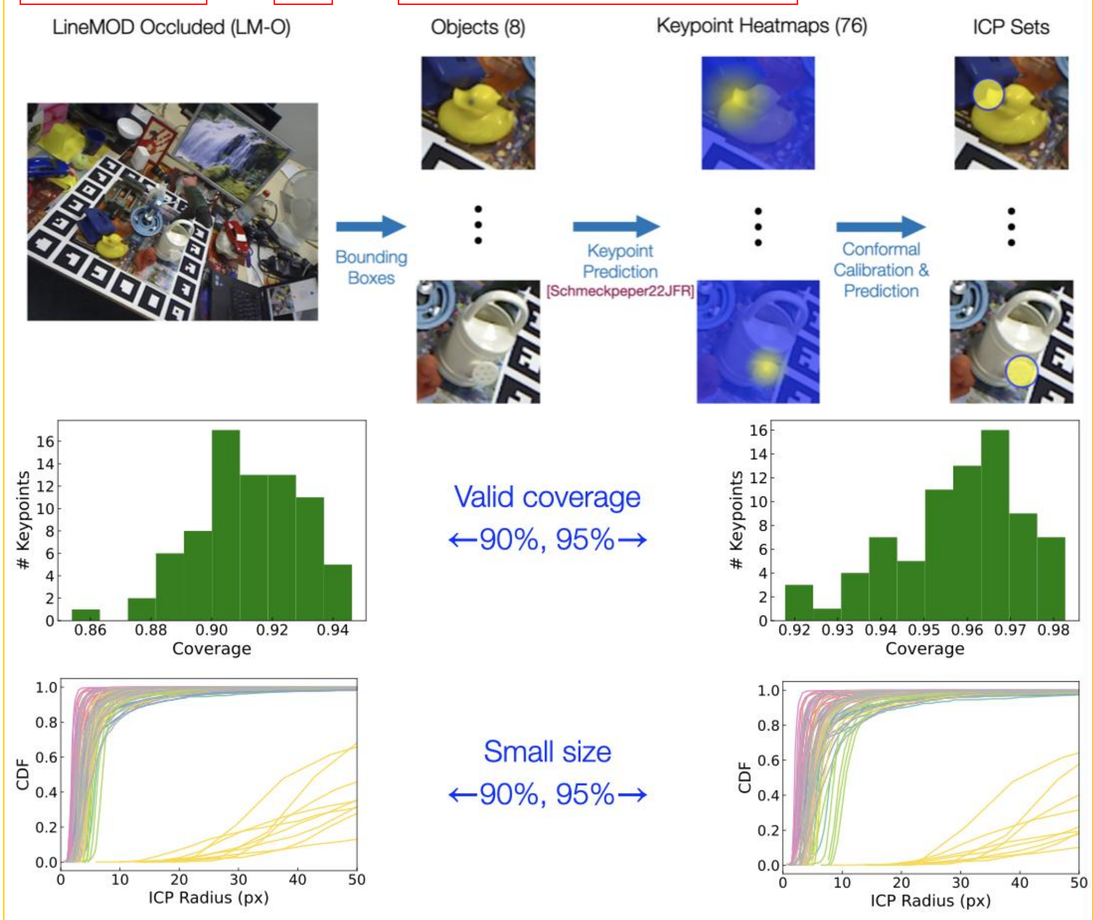
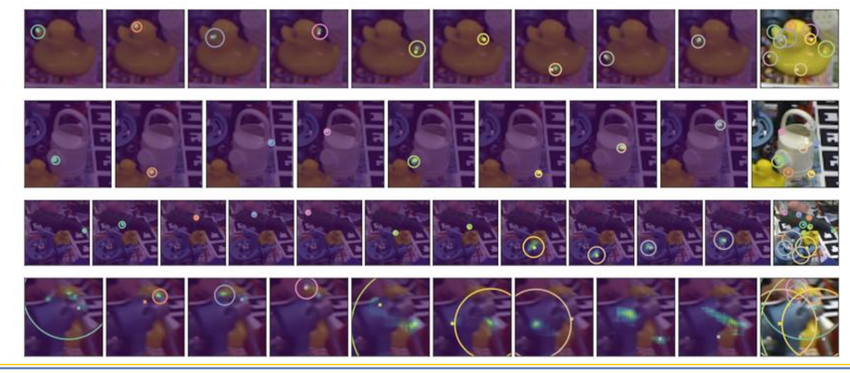


Input:
Heuristic prediction (e.g., heatmap of the keypoint location)

Requirement #2:
See paper for details!

Guarantee:
Cover the true keypoints with a user-specified probability (e.g., 90%)

Experiments: works out of the box!

Setup: 200 images for calibration, 1214 images for testing

User: give me 90% and 95% coverage of the true keypoints